Table 1. Octinoxate: MM/GBSA binding free energy, ΔG (kcal/mol), and ligand residence (%) from 20 ns MD, started from each of three poses in each of three models' UNFORCED co-folds, plus the pocket-guided OpenFold3 lane.

peptide	pose	Boltz / BoltzGen co-fold	OpenFold3 co-fold	RoseTTAFold3 co-fold	OpenFold3 pocket-guided
orig	co-fold	−7.24 (29.9%) •	−5.90 (7.4%) ↑	**−19.51 (99.9%) ↑**	−8.06 (21.1%) ↓
	Vina p1	−2.81 (0.6%) ↑	−9.65 (50.2%) ↑	**−18.71 (89.2%) ↑**	−12.27 (51.0%) •
	GNINA pick (p1/p1/p5/p4)	see Vina p1	see Vina p1	−11.10 (71.0%) ↑	*running*
s2_esm2_control	co-fold	−16.25 (59.6%) •	−10.21 (33.9%) ↓	−14.98 (57.2%) ↓	−16.28 (100.0%) •
	Vina p1	−15.49 (0.0%) ↓	−17.27 (95.2%) ↓	−15.86 (78.9%) •	*running*
	GNINA pick (p6/p5/p4/p8)	−17.09 (50.3%) •	−6.84 (9.4%) ↓	−15.41 (73.7%) ↓	*running*
s3_esm2	co-fold	−15.70 (29.4%) ↑	−13.76 (84.8%) ↑	−16.18 (88.2%) ↑	*running*
	Vina p1	−9.65 (2.4%) ↓	−14.53 (12.6%) •	**−20.39 (99.4%) •**	*running*
	GNINA pick (p7/p8/p3/p6)	−16.64 (0.1%) •	−12.79 (6.9%) ↑	**−20.79 (99.3%) ↑**	*running*
s3_orig	co-fold	−13.37 (48.0%) ↑	−10.10 (44.8%) ↑	−12.10 (68.9%) ↑	−9.78 (43.0%) ↑
	Vina p1	−9.41 (36.8%) •	−8.92 (25.2%) •	**−19.16 (90.0%) ↓**	−11.50 (58.0%) ↓
	GNINA pick (p9/p6/p5/p9)	−11.51 (37.5%) ↑	−11.17 (22.9%) ↑	−8.43 (7.2%) ↑	*running*
shuffle_control	co-fold	−15.13 (75.9%) ↑	−9.39 (25.7%) ↓	−11.13 (38.7%) •	—
	Vina p1	−16.15 (57.8%)	−10.89 (23.3%) •	−5.37 (21.8%) ↑	—
	GNINA pick (p7/p8/p7/—)	−8.40 (22.4%) ↑	−7.78 (12.6%) ↑	−13.86 (60.3%) ↓	—
shuffle_control_esm0	co-fold	−14.32 (80.2%) ↓	−15.62 (92.5%) ↓	**−21.53 (99.3%) •**	—
	Vina p1	−15.50 (76.1%) ↑	−12.27 (1.9%) ↓	−12.47 (44.1%) •	—
	GNINA pick (p9/p3/p3/—)	−14.76 (90.5%) •	**−18.19 (98.5%) •**	−12.03 (36.9%) ↑	—
bg33_1	co-fold	−11.31 (14.5%) ↓	−14.90 (56.4%) ↓	−12.43 (68.3%) ↓	—
	Vina p1	−13.47 (44.7%) ↓	−13.16 (22.6%) •	−14.02 (60.3%) ↓	—
	GNINA pick (p2/p6/p8/—)	−12.55 (42.5%) ↓	−12.93 (44.5%) ↑	−11.78 (62.6%) •	—
bg33_2	co-fold	−8.02 (50.9%) ↑	−11.58 (70.2%) •	−11.52 (58.7%) ↓	—
	Vina p1	−11.82 (43.1%) ↑	−11.55 (14.8%) •	−10.95 (81.5%) •	—
	GNINA pick (p4/p1/p8/—)	−10.44 (57.2%) ↓	see Vina p1	−12.19 (36.7%) •	—
bg33_3	co-fold	−11.86 (41.0%) ↓	−10.10 (14.9%) •	−12.44 (1.9%) •	—
	Vina p1	−10.84 (25.7%) ↑	−14.89 (99.7%) •	−11.01 (0.2%) ↑	—
	GNINA pick (p6/p6/p5/—)	−13.09 (77.6%) ↑	−14.92 (99.6%) •	−11.47 (0.1%) ↑	—
bg33_4	co-fold	**−19.66 (100.0%) ↓**	−17.11 (93.0%) ↓	−16.13 (81.2%) ↑	—
	Vina p1	−14.81 (59.3%) ↑	−16.04 (99.2%) •	**−18.59 (97.6%) ↓**	—
	GNINA pick (p6/p7/p4/—)	−16.53 (97.2%) •	−16.24 (82.8%) ↑	**−19.06 (99.6%) •**	—

Values are ΔG in kcal/mol with residence in parentheses — the share of the whole 20 ns run with the ligand within 10 Å of the peptide. Each peptide carries three sub-rows: the co-fold pose each model produced, Vina p1 (AutoDock rank 1 docked into that same fold), and the GNINA pick (max CNNaffinity over poses ≥1). The pick's pose in each lane is in the row label, in column order; where it is p1 the pick coincides with Vina's rank 1, so no separate leg was ever run and the cell says *see Vina p1* rather than repeating the number. The trailing marker compares the 20 ns ΔG with the ΔG over the first 10 ns: • stable (within 0.5 kcal/mol), ↓ still decreasing (binding strengthening), ↑ still increasing (binding weakening) — a convergence flag on ΔG, not on residence. Bold marks ΔG ≤ −18. Every cell is a finished 20 ns leg; the campaign completed 2026-10-19.

Residence is a centre-of-mass measure, so read it with the fold's shape in mind. It is the share of frames with the ligand's centroid within 10 Å of the peptide's, which penalises an elongated fold: a ligand gripped at one end of a rod is never released and still scores low, because the far half of the peptide drags the centroid away. s3_esm2 co-fold is the case in point — 29.4% residence with 0.15% of frames detached, i.e. essentially permanent atomic contact. Where retention is the question, the release statistics in md_contacts.csv (episode count, longest episode, late share) are the shape-independent measure.

Table 2. Oxybenzone: MM/GBSA binding free energy, ΔG (kcal/mol), and ligand residence (%) from 20 ns MD, started from each of three poses in each of three models' UNFORCED co-folds, plus the pocket-guided OpenFold3 lane.

peptide	pose	Boltz / BoltzGen co-fold	OpenFold3 co-fold	RoseTTAFold3 co-fold	OpenFold3 pocket-guided
ox1_esm1	co-fold	**−26.09 (100.0%) ↓**	−17.95 (100.0%) ↑	−12.12 (35.2%) ↓	−10.87 (95.3%) •
	Vina p1	−10.23 (68.6%) ↑	−10.60 (89.9%) ↑	−7.26 (27.3%) ↑	*running*
	GNINA pick (p4/p2/p4/p8)	−10.11 (56.2%) ↑	−12.79 (66.9%) •	−14.92 (79.8%) ↓	*running*
ox1_orig	co-fold	−8.34 (63.8%) ↑	−3.96 (20.5%) ↑	−12.27 (89.1%) ↓	**−18.26 (100.0%) •**
	Vina p1	−9.34 (61.6%) •	−4.93 (25.5%) ↑	−6.40 (37.5%) ↑	*running*
	GNINA pick (p6/p4/p1/p4)	−4.79 (17.0%) ↑	−5.84 (33.0%) ↓	see Vina p1	*running*
ox2_esm1	co-fold	−12.70 (71.0%) ↑	−16.68 (93.2%) ↑	−10.90 (90.4%) ↑	−9.21 (31.8%) •
	Vina p1	−4.80 (21.2%) ↑	−16.87 (93.9%) ↓	−11.47 (79.1%) ↓	*running*
	GNINA pick (p9/p3/p5/p6)	−11.09 (86.2%) ↓	−10.58 (77.1%) ↑	−9.27 (62.4%) ↓	*running*
ox2_orig	co-fold	−8.51 (43.9%) ↑	−7.23 (49.5%) ↑	−4.95 (19.6%) ↑	*running*
	Vina p1	−6.71 (44.0%) •	−9.05 (47.2%) ↑	−4.22 (15.5%) •	*running*
	GNINA pick (p9/p2/p7/p5)	−3.50 (17.0%) ↑	−5.43 (37.1%) ↑	−4.11 (8.2%) •	*running*
ox2_shuffle	co-fold	−5.30 (7.1%) •	−2.98 (14.8%) ↑	−13.35 (51.0%) •	—
	Vina p1	−5.86 (26.2%) •	−5.91 (27.3%) ↓	−8.60 (72.1%) ↓	—
	GNINA pick (p4/p1/p2/—)	−2.09 (1.2%) •	see Vina p1	−9.29 (32.0%) ↓	—
ox2_shuffle_esm0	co-fold	−12.04 (94.7%) •	−0.92 (1.1%) •	−11.44 (32.6%) •	—
	Vina p1	−12.78 (97.4%) •	−9.44 (60.4%) ↓	−9.18 (54.8%) ↑	—
	GNINA pick (p9/p6/p1/—)	−11.87 (95.5%) •	−6.84 (56.3%) ↑	see Vina p1	—
ox3_orig	co-fold	−10.13 (72.0%) ↑	−2.06 (2.4%) ↓	−7.41 (45.6%) ↓	*running*
	Vina p1	−14.33 (97.9%) ↓	−4.12 (11.8%) ↑	−4.32 (19.5%) •	*running*
	GNINA pick (p6/p4/p8/p1)	−15.44 (100.0%) ↑	−5.99 (33.2%) •	−4.90 (10.0%) ↑	see Vina p1
bgox31_2	co-fold	−10.89 (33.7%) ↑	−7.45 (65.9%) ↑	−9.06 (84.6%) ↓	—
	Vina p1	−2.11 (8.8%) ↑	−5.85 (10.0%) ↓	−7.36 (47.9%) ↑	—
	GNINA pick (p6/p1/p3/—)	−8.50 (2.6%) •	see Vina p1	−9.57 (52.9%) •	—
bgox31_3	co-fold	−15.44 (99.7%) ↑	−5.82 (15.8%) ↑	−14.36 (99.2%) •	—
	Vina p1	**−22.23 (100.0%) ↓**	−10.99 (68.5%) ↑	−13.48 (95.7%) ↑	—
	GNINA pick (p3/p4/p1/—)	−17.52 (89.9%) ↑	−12.48 (88.9%) •	see Vina p1	—
bgox31_4	co-fold	−14.05 (100.0%) ↑	−5.74 (17.9%) ↑	−11.02 (72.0%) •	—
	Vina p1	−9.20 (36.5%) ↑	−10.39 (49.9%) ↑	−15.86 (99.9%) •	—
	GNINA pick (p6/p7/p3/—)	−4.31 (4.9%) ↑	−6.71 (11.3%) ↓	−13.91 (100.0%) •	—
bgox31_5	co-fold	−16.78 (100.0%) •	−11.55 (70.7%) ↓	−14.69 (96.4%) ↑	—
	Vina p1	−12.81 (70.3%) •	−13.72 (93.4%) •	−14.93 (99.8%) •	—
	GNINA pick (p2/p7/p8/—)	−14.28 (99.5%) •	−14.84 (99.5%) ↑	−14.76 (94.9%) •	—

Values are ΔG in kcal/mol with residence in parentheses — the share of the whole 20 ns run with the ligand within 10 Å of the peptide. Each peptide carries three sub-rows: the co-fold pose each model produced, Vina p1 (AutoDock rank 1 docked into that same fold), and the GNINA pick (max CNNaffinity over poses ≥1). The pick's pose in each lane is in the row label, in column order; where it is p1 the pick coincides with Vina's rank 1, so no separate leg was ever run and the cell says *see Vina p1* rather than repeating the number. The trailing marker compares the 20 ns ΔG with the ΔG over the first 10 ns: • stable (within 0.5 kcal/mol), ↓ still decreasing (binding strengthening), ↑ still increasing (binding weakening) — a convergence flag on ΔG, not on residence. Bold marks ΔG ≤ −18. Every cell is a finished 20 ns leg; the campaign completed 2026-10-19.

Residence is a centre-of-mass measure, so read it with the fold's shape in mind. It is the share of frames with the ligand's centroid within 10 Å of the peptide's, which penalises an elongated fold: a ligand gripped at one end of a rod is never released and still scores low, because the far half of the peptide drags the centroid away. s3_esm2 co-fold is the case in point — 29.4% residence with 0.15% of frames detached, i.e. essentially permanent atomic contact. Where retention is the question, the release statistics in md_contacts.csv (episode count, longest episode, late share) are the shape-independent measure.

Table 3. The strict comparison, static half: one sequence, four co-folding treatments.

peptide	Boltz forced	Boltz unforced	OF3 unforced	RF3 unforced
orig_f12 (oct)	0.960 / 4.2 Å	**0.495 / 20.9 Å**	0.680 / 9.4 Å	**0.800 / 5.8 Å**
s3_esm2_f4 (oct)	0.795 / 7.0 Å	0.360 / 13.6 Å	**0.580 / 7.3 Å**	0.460 / 9.3 Å
s3_orig_f12 (oct)	0.965 / 3.6 Å	**0.735 / 9.5 Å**	0.430 / 10.4 Å	0.720 / 8.5 Å
ox1_esm1_f8	0.940 / 4.1 Å	**0.765 / 5.9 Å**	0.580 / 6.1 Å	0.530 / 9.3 Å
ox1_orig_f12	0.995 / 1.2 Å	0.520 / 7.2 Å	**0.810 / 5.7 Å**	0.550 / 6.1 Å
ox2_esm1_f8	0.665 / 6.4 Å	0.610 / 7.6 Å	**0.745 / 5.3 Å**	0.570 / 6.4 Å
ox2_orig_f8	0.975 / 2.7 Å	0.535 / 9.0 Å	**0.655 / 7.5 Å**	0.595 / 6.2 Å
ox3_orig_f8	0.995 / 2.1 Å	**0.615 / 5.8 Å**	0.540 / 10.3 Å	0.465 / 11.4 Å
mean	**0.911 / 3.9 Å**	0.579 / 9.9 Å	0.628 / 7.8 Å	0.586 / 7.9 Å

Each cell is enclosed fraction / ligand-centroid separation. Bold marks the best enclosure among the three *unforced* lanes in that row (Boltz forced is excluded from that contest — it is the treatment under test, not a competitor).

What the strict comparison says, and it is not what Tables 1 and 2 imply. The three unforced lanes are indistinguishable on this measure: mean enclosure 0.579 (Boltz), 0.628 (OF3), 0.586 (RF3), with per-peptide wins splitting 3 / 4 / 1 across eight sequences and every lane's range overlapping the others' almost completely. Forced Boltz is in a different regime from all three — 0.911 enclosed at 3.9 Å, against 0.58–0.63 at 7.8–9.9 Å — and no unforced lane's best cell (0.810) reaches forced Boltz's mean. So the Boltz column's advantage in Tables 1 and 2 is attributable to the contact hints, not to Boltz being the better co-folder: strip the hints and it places the ligand no better than the other two, and on octinoxate's orig_f12 it is the worst of the three by a wide margin (0.495 at 20.9 Å, the furthest starting pose in the whole campaign).